

*Notebook on*  
**Computer Network**

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# Subnetting

192.168.10.5/28:    11000000 . 10101000 . 00001010 . **1111** **0000**  
                                Network bits                          SB      HB

- ## 1.2 Subnetting Example

**Solution:**

IP range : 10111110 . 10101000 . 000000 00 . 00000000 → 190.168.0.0  
10111110 . 10101000 . 000000 11 . 11111111 → 190.168.3.255

### Network-3:

Broadcast address : 10111110 . 10101000 . 00000001 . 11111111 = 190 . 168 . 1 . 255

### Network-1:

Broadcast address : 10111110 . 10101000 . 00000010 . 01111111 = 190 . 168 . 2 . 127

**Network-2:**

Hosts required : 55

Nearest power of 2 :  $2^6 = 64 \rightarrow 32 - 6 = 26$

Subnet address : 10111110 . 10101000 . 00000010 . 10000000 ← 190.168.2.128/26

Subnet Mask : 11111111 . 11111111 . 11111111 . 11000000 = 255 . 255 . 255 . 192

Network address : 10111110 . 10101000 . 00000010 . 10000000 = 190 . 168 . 2 . 128

Broadcast address : 10111110 . 10101000 . 00000010 . 10111111 = 190 . 168 . 2 . 191

## Chapter 2

# Application Layer

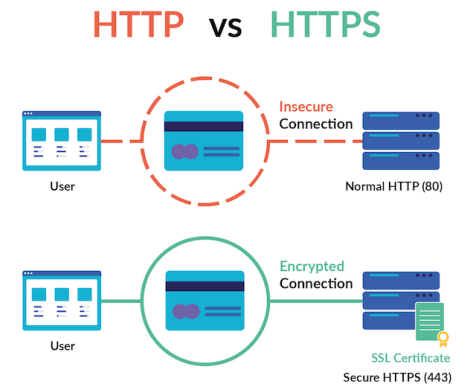
### 2.1 Application Layer Protocols

#### HTTP (Hypertext Transfer Protocol)

Used for transferring hypertext requests and information between web browsers and web servers. It forms the basis of communication for the World Wide Web.

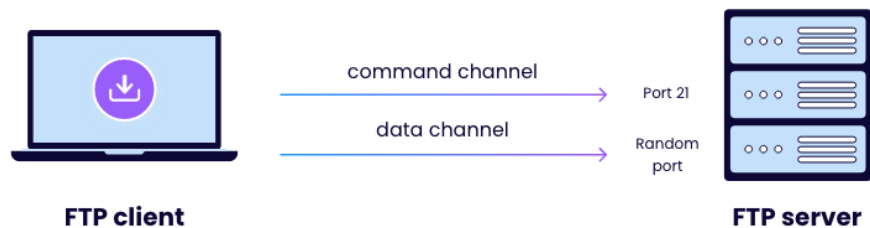
#### HTTPS (Hypertext Transfer Protocol Secure)

A secure version of HTTP that incorporates encryption and authentication mechanisms provided by SSL/TLS protocols for secure communication over the internet.



#### FTP (File Transfer Protocol)

Enables the transfer of files between a client and a server on a network. It allows users to upload, download, list, rename, and delete files on remote servers.

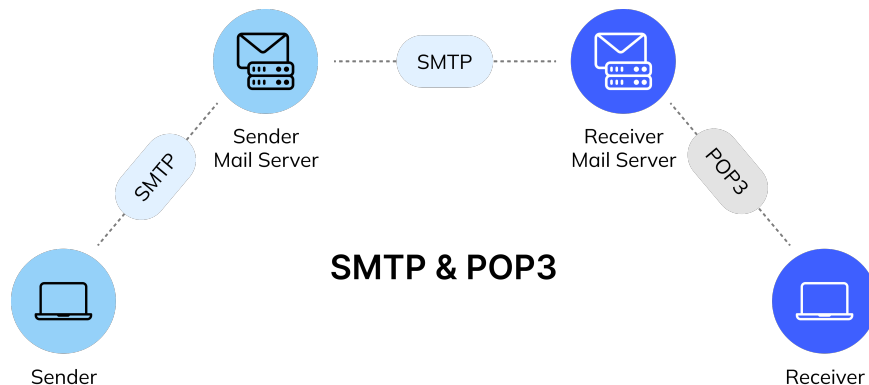


#### SMTP (Simple Mail Transfer Protocol):

Used for sending email messages between email servers. It defines how email messages are transferred and routed across networks.

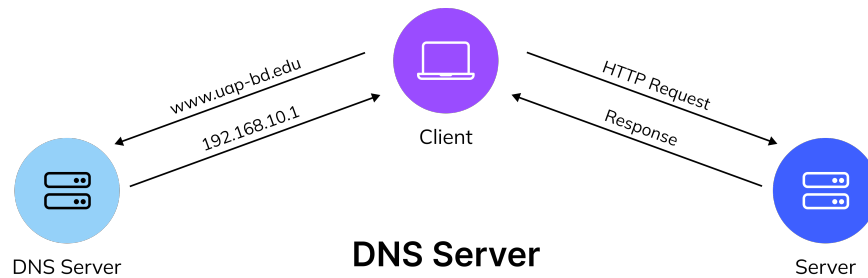
#### POP3 (Post Office Protocol version 3):

Allows email clients to retrieve email messages from a remote server. It is commonly used for downloading emails to the client's device.



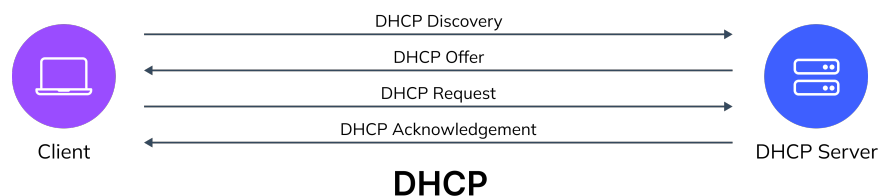
## 2.2 DNS (Domain Name System):

Resolves domain names to IP addresses and vice versa. It translates human-readable domain names (e.g., www.example.com) into numerical IP addresses used by computers to communicate over a network.



## 2.3 DHCP (Dynamic Host Configuration Protocol):

Automatically assigns IP addresses and network configuration parameters to devices on a network. It simplifies the process of network configuration and management for both users and administrators. DHCP is an application layer protocol.



## 2.4 Comparison of Common Application Layer Protocols

| Protocol | Stateful/Stateless                            | Transport Protocol | Persistent     | Port No.                | Data Transfer |
|----------|-----------------------------------------------|--------------------|----------------|-------------------------|---------------|
| DNS      | Stateless (usually)                           | UDP                | Non-persistent | 53                      | In-band       |
| HTTP     | Stateless (default),<br>Stateful with cookies | TCP                | Persistent     | 80, 443                 | In-band       |
| SMTP     | Stateful                                      | TCP                | Persistent     | 25                      | In-band       |
| POP3     | Stateful                                      | TCP                | Persistent     | 110                     | In-band       |
| FTP      | Stateful                                      | TCP                | Persistent     | 21 (control), 20 (data) | Out-of-band   |

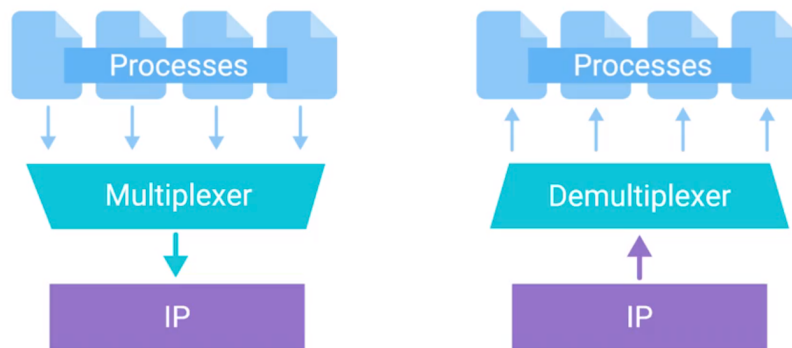
## Chapter 3

# Transport Layer

### 3.1 Multiplexing and Demultiplexing

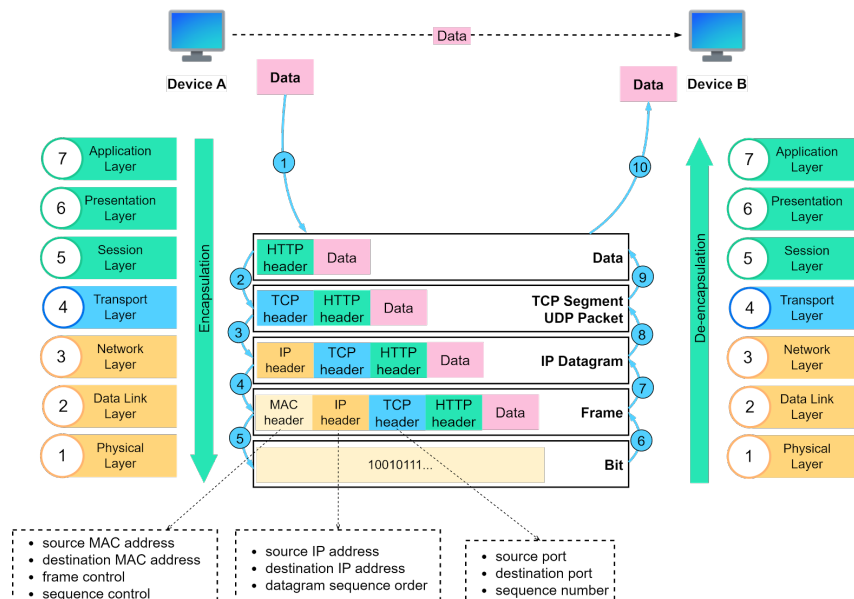
Multiplexing is the process at the sender's side where multiple applications or processes use the same network connection simultaneously. It **combines data from multiple sources into a single stream** and tags each segment with relevant identifiers (like port numbers) so the receiver can understand which segment belongs to which application.

Demultiplexing is the process at the receiver's side where the **combined data stream is split back into separate application data**, based on the identifiers (typically, destination IP address and port number). This ensures each segment is delivered to the correct application.



### 3.2 Encapsulation and Decapsulation

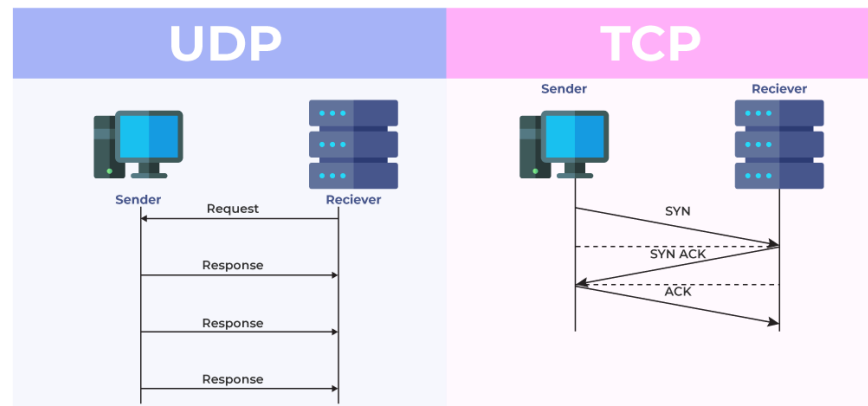
Encapsulation is the process of adding header and trailer information to a data packet at each layer of the OSI model, while decapsulation is the reverse process of removing these headers and trailers at the receiving end. This structured approach ensures data transmission and reception across a network.





### 3.3 TCP and UDP

| Transmission Control Protocol(TCP)   | User Datagram Protocol (UDP) |
|--------------------------------------|------------------------------|
| connection oriented protocol         | connectionless protocol      |
| reliable                             | not reliable                 |
| slower                               | faster                       |
| An acknowledgment segment is present | No acknowledgment segment    |
| possible retransmission              | no retransmission            |
| 20-60 bytes variable length header   | 8 bytes fixed length header  |



### 3.4 SSL and TLS

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are cryptographic protocols designed to provide secure communication over a network. SSL was the original protocol developed by Netscape, but it has known vulnerabilities and is now considered obsolete.

TLS is the modern, more secure successor to SSL. It offers better encryption algorithms, improved key exchange mechanisms, and stronger message authentication. TLS ensures confidentiality, integrity, and authentication in data transmission, and is widely used in securing web traffic (HTTPS), emails, and VoIP.

### 3.5 Practice Problems

#### 1. TCP Throughput Calculation

- **Question:** A TCP connection has a window size of 65,535 bytes and a round-trip time (RTT) of 100 ms. What is the maximum achievable throughput?
- **Solution:**

$$\begin{aligned}\text{Throughput} &= \frac{\text{Window Size}}{\text{RTT}} \\ &= \frac{65535 \text{ bytes}}{0.1 \text{ sec}} \\ &= 655,350 \text{ bytes/sec} \approx 655.35 \text{ KBps}\end{aligned}$$

## 2. UDP vs TCP Header Overhead

- **Question:** If you send a 1000-byte message using TCP and UDP, what is the actual data size transmitted (including headers)?
- **Solution:**
  - **TCP:** Header = 20 bytes, IP Header = 20 bytes  
Total = 1000 + 20 + 20 = 1040 bytes
  - **UDP:** Header = 8 bytes, IP Header = 20 bytes  
Total = 1000 + 8 + 20 = 1028 bytes

## 3. Segment Count for a File

- **Question:** A file of size 10 MB is being sent using TCP, with each segment carrying 1460 bytes of data. How many segments are needed?
- **Solution:**

$$\begin{aligned}\text{Segments} &= \left\lceil \frac{10 \times 1024 \times 1024}{1460} \right\rceil \\ &= \left\lceil \frac{10485760}{1460} \right\rceil \\ &= 7184 \text{ segments}\end{aligned}$$

## 4. Round-Trip Delay and Acknowledgment

- **Question:** A sender is transmitting packets of 1000 bytes over a TCP connection. The RTT is 200 ms, and the receiver sends an acknowledgment every 2 packets. What is the average data rate?
- **Solution:**
  - Total data acknowledged per RTT =  $2 \times 1000 = 2000$  bytes
  - RTT = 0.2 sec

$$\begin{aligned}\text{Throughput} &= \frac{2000}{0.2} \\ &= 10,000 \text{ bytes/sec} \\ &= 10 \text{ KBps}\end{aligned}$$

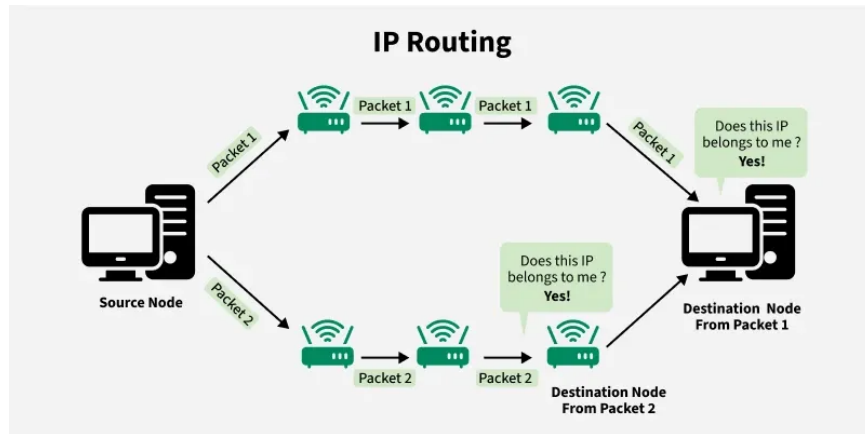
## Chapter 4

# Network Layer

### 4.1 Routing

**Routing:** Routing is the process of selecting a path for data to travel from a source to a destination across interconnected networks. It ensures data packets are delivered efficiently through the best possible route.

**Router:** A router is a networking device that connects multiple networks together and forwards data packets between them based on their destination IP address. It makes routing decisions using routing tables and protocols.

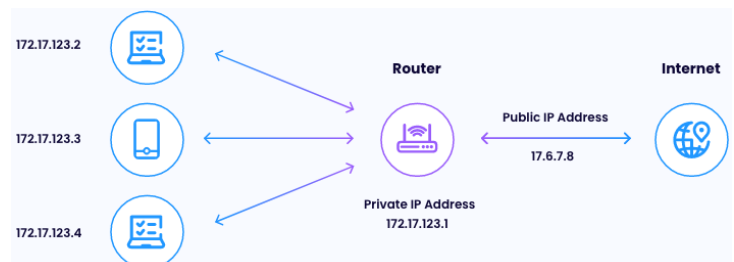


### 4.2 Difference between Static and Dynamic Routing

| Static Routing                              | Dynamic Routing                                             |
|---------------------------------------------|-------------------------------------------------------------|
| Routes are user-defined.                    | Automatically adjusted based on network conditions.         |
| Does not use complex routing algorithms.    | Uses complex routing algorithms to determine the best path. |
| Provides high security.                     | Less secure.                                                |
| Does not require additional resources.      | Require additional resources.                               |
| Failure of the link disrupts the rerouting. | Failure of the link doesn't disrupts the rerouting.         |
| Requires less bandwidth.                    | Requires more bandwidth.                                    |

### 4.3 Network Address Translation (NAT)

Network Address Translation (NAT) is a process in which one or more local IP addresses are translated into one or more Global IP addresses and vice versa to provide Internet access to the local hosts. It also does the translation of port numbers, i.e., masks the port number of the host with another port number in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.



## 4.4 Main Routing Protocols

| RIP (Routing Information Protocol)                                                                               | OSPF (Open Shortest Path First)                                                                                |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| A distance-vector routing protocol that uses hop count (max 15 hops) to determine the best path between routers. | A link-state routing protocol that uses a map of the network and calculates the shortest path between routers. |
| RIP uses Bellman-Ford algorithm                                                                                  | OSPF uses Dijkstra algorithm.                                                                                  |
| Broadcasts entire routing table to all neighbors every 30s. This is also known as Routing on rumors.             | Multicasts only changes using LSAs only when there is a change in the network topology.                        |
| RIP v1: Classful, RIP v2: Classless                                                                              | Classless (supports VLSM and CIDR)                                                                             |
| Suitable for small networks.                                                                                     | Suitable for larger networks.                                                                                  |
| Slow convergence.                                                                                                | Faster convergence.                                                                                            |
| Uses UDP (Port 520)                                                                                              | Runs directly over IP (Protocol number 89)                                                                     |

**EIGRP (Enhanced Interior Gateway Routing Protocol):** It is a hybrid protocol that combines features of distance-vector and link-state.

**BGP (Border Gateway Protocol):** It is a path-vector protocol that is used for routing between different autonomous systems on the internet.

**Distance Vector Routing:** All the nodes that are a part of the network advertise their routing table to their adjacent nodes (nodes that are directly connected) at regular intervals. Algorithm used: Bellman Ford Algorithm to find the shortest path.

## 4.5 Difference Between IPv4 and IPv6

| Property              | IPv4                                         | IPv6                                       |
|-----------------------|----------------------------------------------|--------------------------------------------|
| Address Length        | 32 bits (decimal representation)             | 128 bits (hexadecimal representation)      |
| Address configuration | Manual and DHCP                              | Auto and renumbering                       |
| Connection integrity  | unachievable end to end connection integrity | achievable end to end connection integrity |
| Encryption & Auth     | not provided by default                      | provided by default                        |
| Security              | dependent on application                     | built-in security features (IPSec)         |
| Header size           | 20-60 bytes (variable)                       | 40 bytes (fixed)                           |
| Subnetting            | Supports VLSM                                | Doesn't supports VLSM                      |
| Example               | 66.94.29.13                                  | 2001:0000:3238:DFE1:0063:0000:0000:A0F8    |

## Chapter 5

# Error Control

### 5.1 Checksum

Checksum is an error-detection technique used to ensure data integrity during transmission. It helps detect errors by appending a checksum value derived from the data. The receiver verifies integrity by checking if the total (including checksum) equals all 1s.

#### Advantages:

- Simple to compute and implement in hardware/software.
- Efficient for detecting common transmission errors.
- Low computational overhead, and widely used in protocols (TCP/IP, UDP)

### 5.2 Practice Problems

**Question:** Consider, the data unit to be transferred is 10010011 10010011 10011000 01001101. Checksum is an efficient error detection method to detect whether there is an error. Suppose, 8 bit checksum is used in this case. Now use this method on the data unit given above and answer the following: Briefly explain the above-mentioned method and why it is a good choice. Calculate the checksum and the exact data that will be transmitted.

#### Solution:

In sender side,

$$\begin{array}{r} 10010011 \\ 10010011 \\ 10011000 \\ + 01001101 \\ \hline 10 \ 00001011 \\ \phantom{10} 10 \\ \hline 00001101 \end{array}$$

∴ Checksum is: 11110010 (1's compliment)

∴ The data will be transmitted as,

10010011 10010011 10011000 01001101 **11110010**

In receiver side,

$$\begin{array}{r} 10010011 \\ 10010011 \\ 10011000 \\ 01001101 \\ + 11110010 \\ \hline 10 \ 11111101 \\ \phantom{10} 10 \\ \hline 11111111 \end{array}$$

∴ Since the sum is 11111111, which is all 1s, the data is correct.

## Chapter 6

# Cellular Network

### 6.1 Multiple Access

#### Frequency Division Multiple Access (FDMA)

It assigns a unique frequency channel to each user for the duration of the call. A guard band is needed to prevent interference between adjacent channels. Mobile terminals operate with dedicated uplink and downlink frequencies, but since frequency spectrum is limited and valuable, efficient usage is critical—leading to the exploration of technologies like cognitive radio.

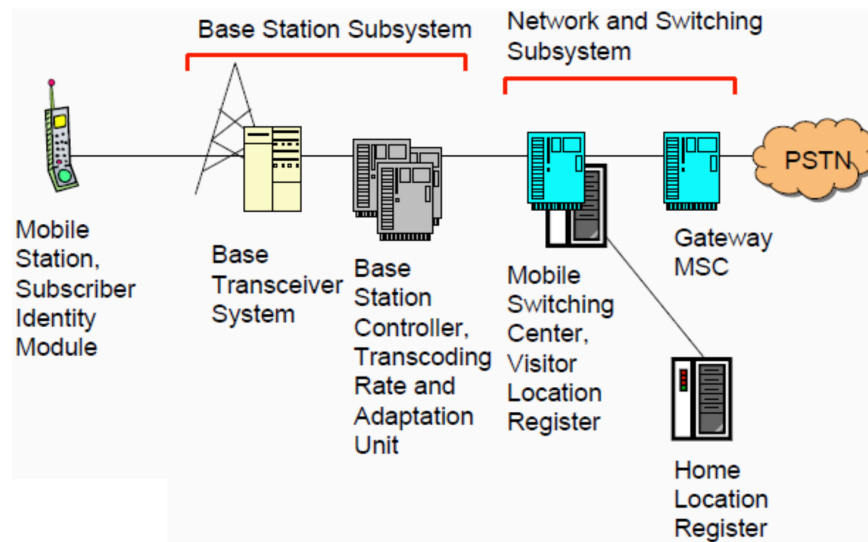
#### Time Division Multiple Access (TDMA)

It divides time into slots, where each user transmits only during their assigned time slot, avoiding contention. Unlike WiFi's CSMA, TDMA ensures no overlap among users. A guard time is necessary to account for different signal arrival times from users at varying distances from the base station.

#### Code Division Multiple Access (CDMA)

It allows all users to share the same frequency band simultaneously by assigning unique orthogonal codes to each. Data is spread over a wider bandwidth using these codes, enabling multiple transmissions to coexist without interference due to code separation.

### 6.2 GSM (Global System for Mobile Communications)



### 6.3 Cellular Network Components

#### Mobile Station (MS)

The mobile station consists of a mobile device (like a smartphone) and a SIM card. The SIM card contains the International Mobile Subscriber Identity (IMSI) and Personal Identification Number (PIN), which uniquely identifies the user on the network.

#### Mobile Switching Center (MSC)

The MSC is a central component in a cellular network that connects calls by establishing a path between the calling and receiving parties. It manages call routing, handovers, mobility management, and collecting billing information.

## Base Station Subsystem (BSS)

The BSS consists of the **Base Transceiver Station (BTS)** and **Base Station Controller (BSC)**. The BTS handles radio communication with mobile devices. It contains the equipment for transmitting and receiving of radio signals (transceivers), antennas, and equipment for encrypting and decrypting communications with the BSC. BSC manages multiple BTSs, controlling radio resources, handovers, and frequency allocation.

### 6.4 Handoff (Handover)

Within a BSS, the BSC, which knows the current radio link configuration (including feedbacks from the MS), prepares an available channel in the new BTS. Then the MS is told to switch over to the new BTS. This is called a hard handoff. In a soft handoff, the MS is connected to two BTSes simultaneously.

## Chapter 7

# Cyber Security

### 7.1 Cyber Security Fundamentals

#### Confidentiality:

Ensures that sensitive information is **accessible only to authorized users**. This prevents unauthorized disclosure of data.

**Techniques includes:** Encryption, access control, authentication.

#### Integrity:

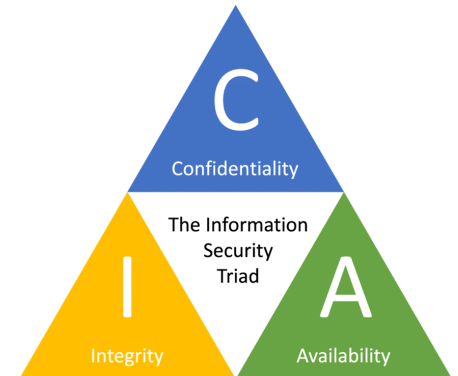
Ensures that data is **accurate and unaltered** during storage, transmission, and processing. This prevents unauthorized modification of data.

**Techniques includes:** Checksums, hash functions, digital signatures.

#### Availability:

Ensures that information and resources are **accessible and usable when needed** by authorized users. This prevents disruptions in service.

**Techniques includes:** data redundancy, protection from DDoS attack, load balancing, implementing firewalls and regular backups.



### 7.2 Cyber Attacks

#### Web Based Attacks

##### Mnemonic:

In Dhaka, Smart People Buy Delicious Dumpling Using Fresh Money

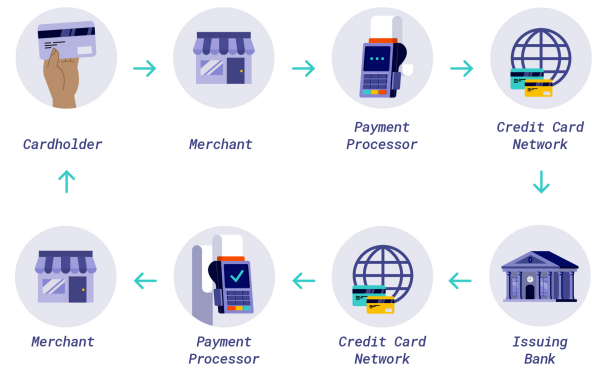
|                           |                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Injection attacks         | : Attacker injects malicious code into a web application and fetch required information.                                       |
| DNS Spoofing              | : Attacker changes the DNS address of a website to redirect the user to a fake website.                                        |
| Session Hijacking         | : By stealing the cookies, an attacker can have access to all of the user data.                                                |
| Phishing                  | : Steals sensitive information like user credentials and credit card number by pretending to be trustworthy sources.           |
| Brute Force               | : A hacking method that uses trial-and-error to crack passwords, credentials, or encryption keys.                              |
| Denial of Service (DOS)   | : Attacker overloads a target server or network with traffic from multiple sources, making it unavailable to legitimate users. |
| Volume-based attacks      | : Its goal is to saturate the bandwidth of the attacked site, and is measured in bit per second.                               |
| Protocol attacks          | : It consumes actual server resources, and is measured in a packet.                                                            |
| Application layer attacks | : Its goal is to crash the web server and is measured in request per second.                                                   |
| Dictionary attack         | : Attacker uses a list of commonly used passwords to guess original one.                                                       |
| URL Interpretation        | : Attacker changes certain parts of a URL, leading to unauthorized access or data exposure.                                    |
| File Inclusion Attack     | : Attacker uses the include feature to access sensitive files or run malicious code on the web server.                         |
| MITM                      | : Attacker to intercepts connection between client and server and acts as a bridge between them.                               |



## 7.3 Credit Card Fraud

With the rise of mobile computing, including M-Commerce and M-Banking, cyber crimes such as credit card fraud are becoming more frequent. The affordability and accessibility of powerful mobile devices make it easier for both consumers and criminals to engage in these systems.

Wireless technology has enabled anytime, anywhere computing, especially for white-collar professionals. This includes wireless credit card processing, which allows businesses to accept payments on the go. While highly convenient and efficient for mobile businesses, it also increases the risk of security breaches, making cybersecurity in mobile environments more critical than ever.

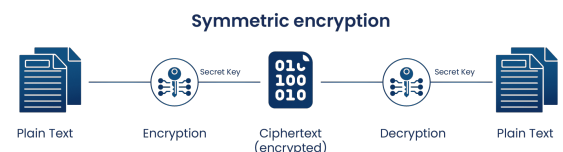


## 7.4 Cryptography

Cryptography is the practice of securing information using codes and algorithms so that only intended recipients can understand it. It relies on mathematical techniques for key generation, encryption, digital signing, and verification to ensure data confidentiality, integrity, and authenticity. The term comes from “crypt” (hidden) and “graphy” (writing).

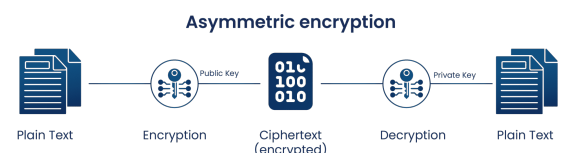
### Symmetric Key Cryptography

Symmetric key cryptography uses a **single shared key for both encryption and decryption**. It's fast and efficient, but requires a secure method to share the key between sender and receiver. Common examples include Data Encryption Systems (DES) and Advanced Encryption Systems (AES).



### Asymmetric Key Cryptography

Asymmetric key cryptography uses a **pair of keys—a public key for encryption and a private key for decryption**. While the public key is openly shared, only the intended receiver can decrypt the message using their private key. A widely used example is the RSA algorithm.



### Hash Functions

There is **no usage of any key** in this algorithm. **Uses a calculated hash value as per the plain text** which makes it impossible for the contents of plain text to be recovered. Many operating systems use hash functions to encrypt passwords.

## 7.5 Types of Cryptographic Algorithms

**AES (Advanced Encryption Standard):** A symmetric key block cipher that encrypts data in 128-bit blocks using 128, 192, or 256-bit keys. It's fast, secure, and the modern standard replacing DES.

**DES (Data Encryption Standard):** An older symmetric algorithm that encrypts 64-bit blocks using a 56-bit key. It's now considered insecure due to its short key length.

**RSA (Rivest-Shamir-Adleman):** A widely used asymmetric algorithm that uses a public key for encryption and a private key for decryption. Based on the difficulty of factoring large integers.

**SHA (Secure Hash Algorithm):** A family of cryptographic hash functions (e.g., SHA-2, SHA-3) used to generate fixed-size hash values from input data to ensure data integrity and authenticity. Even small input changes yield drastically different hashes.